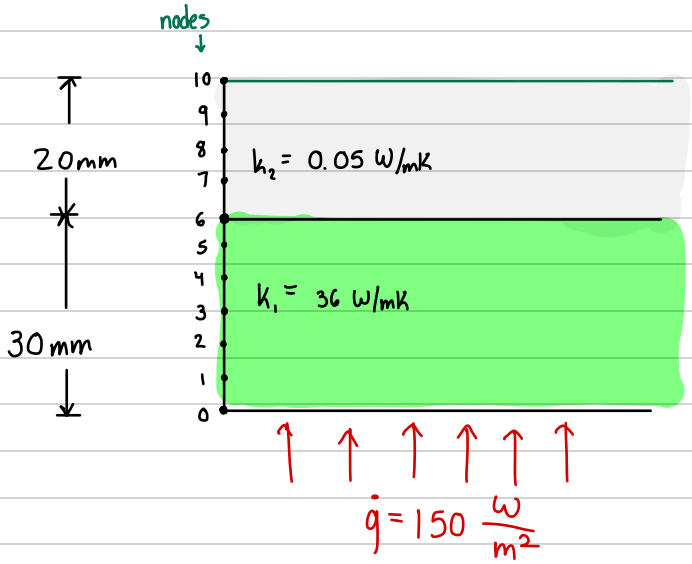
Problem 1

Diagram:

Air

$T_\infty = 15°C, h = 5 W/m^2K$



$\Delta X = 5mm$

Assumptions:

- Steady State
- One dimensional - x
- Uniform properties:
  h = const, k = const, $\dot{q}$ = const, temperatures
- Radiation is negligible
- Mecanismos: Conducción y convección

Procedure:

$$\frac{L}{\Delta X} + 1 = M$$

$$\frac{(20mm + 30mm)}{5mm} + 1 = 11 \text{ nodes}$$

m=10: $\dot{Q} = A_s h (T - T_\infty)$

$\dot{q} = (T_{10} - T_\infty) h$

$\dot{q} = \frac{\dot{Q}}{A}$ $\quad A\dot{q} = \dot{Q}$

$\dot{q}/h + T_\infty = T_{10}$

$150 \frac{W}{m^2} / 5 \frac{W}{m^2 K} \left(\frac{K}{°C}\right) + 15°C = T_{10}$

$30°C + 15°C = 45°C$

$m = 6$: $150 \frac{W}{m^2} = -36 \frac{W}{mK}\left(\frac{1K}{°C}\right) \cdot \frac{\Delta T_1}{0.03m}$

$\Delta T_1 = -0.125\,°C$

$m = 10$: $150 \frac{W}{m^2} = -0.05 \frac{W}{mK}\left(\frac{1K}{°C}\right) \cdot \frac{\Delta T_2}{0.02m}$

$\Delta T_2 = -60\,°C$

$m = 0$: $\Delta T = \Delta T_1 + \Delta T_2$

$T_{10} - T_0 = \Delta T$

$T_{10} - (\Delta T_1 + \Delta T_2) = T_0$

$45°C - (-0.125°C - 60°C) = 105.125°C$

$T_0 = 105.125\,°C$

Since $T_0 = 105.125\,°C$ and $T_0 > 93°C$, the ASME SB-96 plate is not compliant with the ASME Boiler and Pressure Vessel Code, thus it cannot be operated.

Boundary Conditions

$\Delta T_{k=36\frac{W}{mK}}$

$\dot{q}_0 + k_1 \frac{(T_1 - T_0)}{\Delta X} = 0$

$150 \frac{W}{m^2} + 36 \frac{W}{mK}\left(\frac{1K}{°C}\right) \cdot \frac{(\Delta T)}{0.005m} = 0$

$\dot{Q} = -\frac{kA(T_1 - T_0)}{\Delta X}$

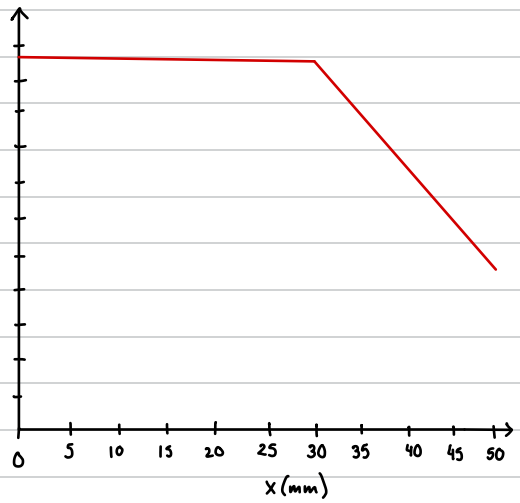
$\dot{q}_0 = -\frac{k_1(T_1 - T_0)}{\Delta X}$

$\Delta T_{k=36\frac{W}{mK}} = -0.02083\,°C$

$150 \frac{W}{m^2} + 0.05 \frac{W}{mK}\left(\frac{1K}{°C}\right) \cdot \frac{(\Delta T)}{0.005m} = 0$

$\Delta T_{k=0.05\frac{W}{mK}} = -15\,°C$

Temperature Profile:



m	X (mm)	T_m (°C)
0	0	105.125
1	5	105.10417
2	10	105.08334
3	15	105.06251
4	20	105.04168
5	25	105.02085
6	30	105.00002
7	35	90
8	40	75
9	45	60
10	50	45

$\Delta T_{k=36} = -0.02083°C$

$\Delta T_{k=0.05} = -15°C$

Complete procedure instead of simply applying ΔT_{hi} from each material. Same results are obtained. For example:

$m = 1$: $T_0 - 2T_1 + T_2 = 0$

$T_1 - T_0 = -0.02083°C$

$T_1 = -0.02083 + T_0$

$m = 2$ $T_0 - 2(-0.02083°C + T_0) + T_2 = 0$

$T_2 = 2(-0.02083 + 105.125°C) - 105.125°C$

$T_2 = 105.083°C$

$\frac{T_2 + T_0}{2} = T_1 \quad \therefore T_1 = \frac{105.083°C + 105.125°C}{2} = 105.104°C$

Lowest h value so allow does not exceed 93°C.

$\Delta T = \Delta T_1 + \Delta T_2$

$\Delta T = -0.125°C - 60°C$

$\Delta T = -60.125°C$

$T_{10} - (\Delta T_1 + \Delta T_2) = T_0$

$T_{10} - (-60.125°C) < 93°C$

$T_{10} < 32.875°C$

$\dot{q} = (T_{10} - T_\infty)h$

$\dot{q}/h + T_\infty = T_{10}$

$\frac{\dot{q}}{h} + T_\infty < 32.875°C$

$\frac{\dot{q}}{h} < 32.875°C - T_\infty$

$150 \frac{W}{m^2} / (32.875°C - 15°C) < h$

$8.39161 \frac{W}{m^2 °C} < h$

$h > 8.39161 \frac{W}{m^2 K}$